

Downstream analysis of single-cell RNA-seq data

We've talked about quantification but, most work is focused on “downstream” analyses

- Denoising : removal of “empty” droplets, detection of “doublets”
- Imputation : detection and inference of “dropouts”*
- Clustering and cell-type identification : Which cells are of the same “type”? What types of cells are present in my experiment?
- Pseudotime analysis: What expression changes are important / determinative in development? What are developmental branchpoints? What is the Waddington landscape?
- Dimensionality reduction for visualization : tSNE and UMAP are currently most popular. How do we visualize 10^3 — 10^6 cells?
- Foundation Models : AI for single-cell.

* “dropout” is a somewhat contentious term / idea — really just the implications of finite sampling from a limited population

Open Problems

 Open Problems

DatasetsBenchmarksEventsAboutDocumentation

 Sign In

nature biotechnology

Defining and benchmarking open problems in single-cell analysis.

Want to know more?

Single-cell analysis is *hard*. Open Problems aims to define and assess those challenges. Our paper describing Open Problems, our structure and philosophy is now out in Nature Biotechnology! Follow the link below to find out more about how we work and what we aim to achieve.

Read more ↗



Benchmarking formalized challenges in single-cell analysis

Advances in microfluidic technology have transformed single-cell analysis, allowing for unprecedented data collection. However, these complex datasets require more than standard statistical techniques.

To fully harness the potential of single-cell biology, we need new methods of data analysis. Our goal is to formalize key challenges and create a community-driven platform to benchmark and advance these methods.

openproblems.bio

Open Problems

Denoising

v1.0.0

Info

3 datasets

11 methods

Repository

MIT

2 control methods

2 metrics

Issues

Removing noise in sparse single-cell RNA-sequencing count data

Task info

Method info

Metric info

Dataset info

Results

Single-cell RNA-Seq protocols only detect a fraction of the mRNA molecules present in each cell. As a result, the measurements (UMI counts) observed for each gene and each cell are associated with generally high levels of technical noise ([Grün et al., 2014](#)). Denoising describes the task of estimating the true expression level of each gene in each cell. In the single-cell literature, this task is also referred to as *imputation*, a term which is typically used for missing data problems in statistics. Similar to the use of the terms "dropout", "missing data", and "technical zeros", this terminology can create confusion about the underlying measurement process ([Sarkar and Stephens, 2021](#)).

A key challenge in evaluating denoising methods is the general lack of a ground truth. A recent benchmark study ([Hou et al., 2020](#)) relied on flow-sorted datasets, mixture control experiments ([Tian et al., 2019](#)), and comparisons with bulk RNA-Seq data. Since each of these approaches suffers from specific limitations, it is difficult to combine these different approaches into a single quantitative measure of denoising accuracy. Here, we instead rely on an approach termed molecular cross-validation (MCV), which was specifically developed to quantify denoising accuracy in the absence of a ground truth ([Batson et al., 2019](#)). In MCV, the observed molecules in a given scRNA-Seq dataset are first partitioned between a *training* and a *test* dataset. Next, a denoising method is applied to the training dataset. Finally, denoising accuracy is measured by comparing the result to the test dataset. The authors show that both in theory and in practice, the measured denoising accuracy is representative of the accuracy that would be obtained on a ground truth dataset.

Most work has focused on “downstream” analysis

- Denoising : removal of “empty” droplets, detection of “doublets”

Cell Systems

Volume 8, Issue 4, 24 April 2019, Pages 281-291.e9



Article

Scrublet: Computational Identification of Cell Doublets in Single-Cell Transcriptomic Data

Samuel L. Wolock¹, Romain Lopez^{1, 2, 3}, Allon M. Klein^{1, 4}  

 [Show more](#)

<https://doi.org/10.1016/j.cels.2018.11.005>

[Get rights and content](#)

Lun et al. *Genome Biology* (2019) 20:63
<https://doi.org/10.1186/s13059-019-1662-y>

Genome Biology

METHOD

Open Access

EmptyDrops: distinguishing cells from empty droplets in droplet-based single-cell RNA sequencing data



Aaron T. L. Lun^{1*†}, Samantha Riesenfeld^{2†}, Tallulah Andrews^{3†}, The Phuong Dao^{4†}, Tomas Gomes^{3†}, participants in the 1st Human Cell Atlas Jamboree and John C. Marioni^{1,3,5*} 

nature methods

Article

<https://doi.org/10.1038/s41592-023-01943-7>

Unsupervised removal of systematic background noise from droplet-based single-cell experiments using CellBender

Received: 10 July 2022

Accepted: 13 June 2023

Published online: 7 August 2023

Stephen J. Fleming^{1,2}  , Mark D. Chaffin^{2,3} , Alessandro Arduini^{2,8} , Amer-Denis Akkad⁴, Eric Banks¹, John C. Marioni^{5,6} , Anthony A. Philippakis¹, Patrick T. Ellinor^{2,3,7}  & Mehrtash Babadi^{1,2}  

BRIEF REPORT | VOLUME 8, ISSUE 4, P329-337.E4, APRIL 24, 2019

DoubletFinder: Doublet Detection in Single-Cell RNA Sequencing Data Using Artificial Nearest Neighbors

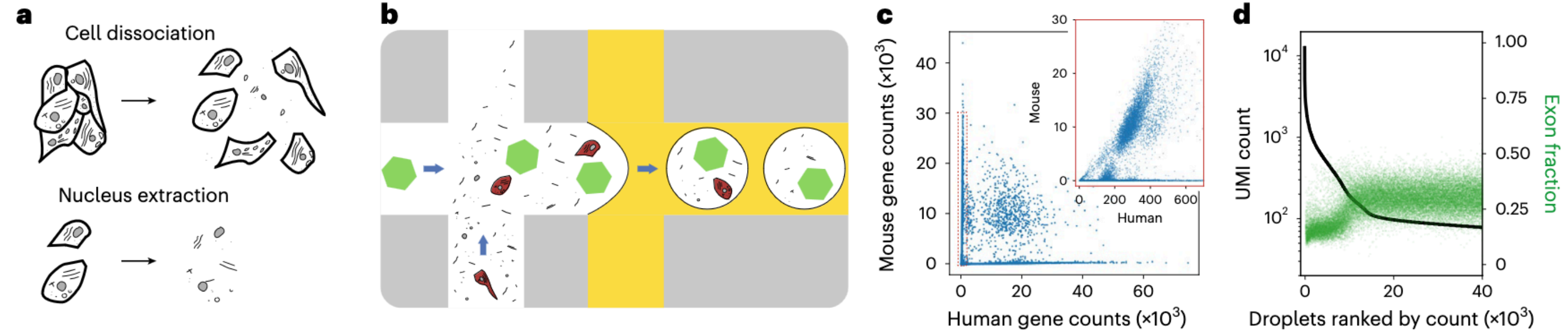
Christopher S. McGinnis • Lyndsay M. Murrow • Zev J. Gartner  ⁴  • [Show footnotes](#)

Published: April 03, 2019 • DOI: <https://doi.org/10.1016/j.cels.2019.03.003> •

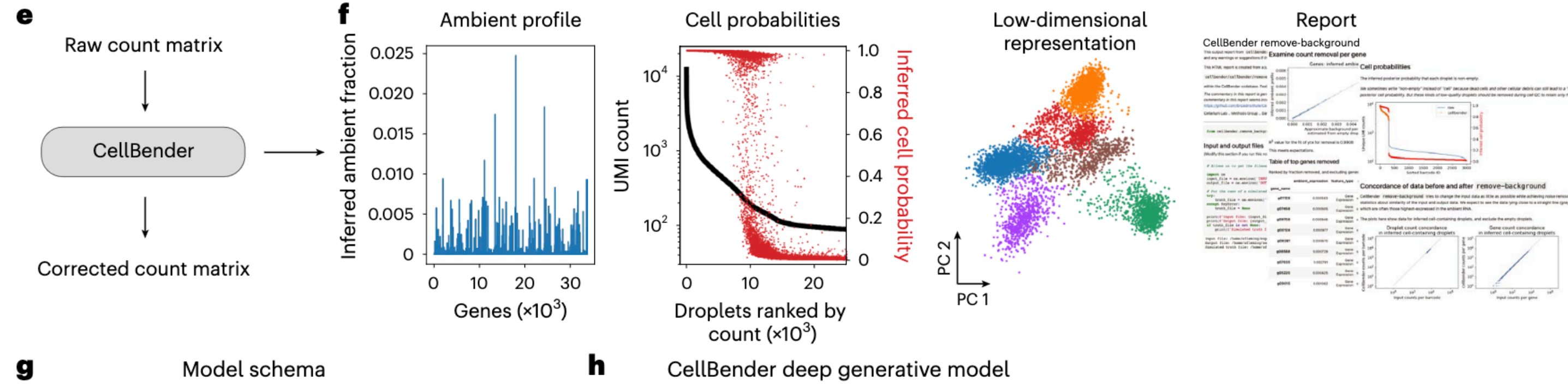


Most work has focused on “downstream” analysis

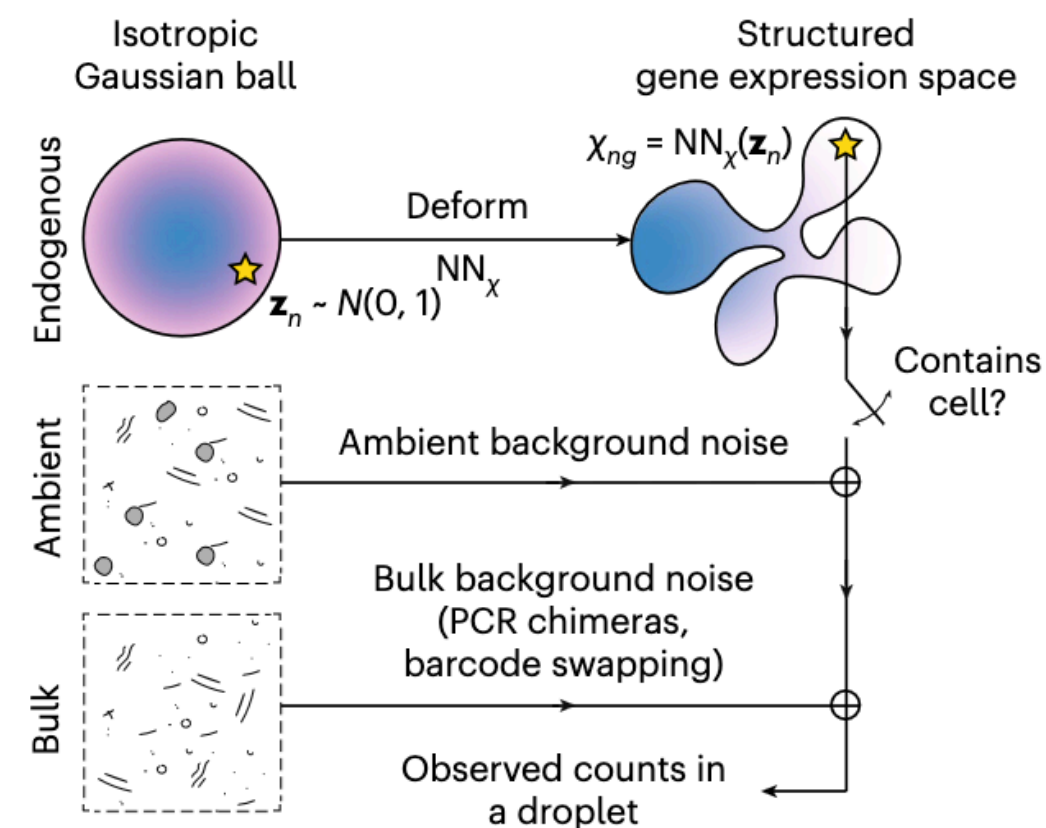
Phenomenology of ambient RNA



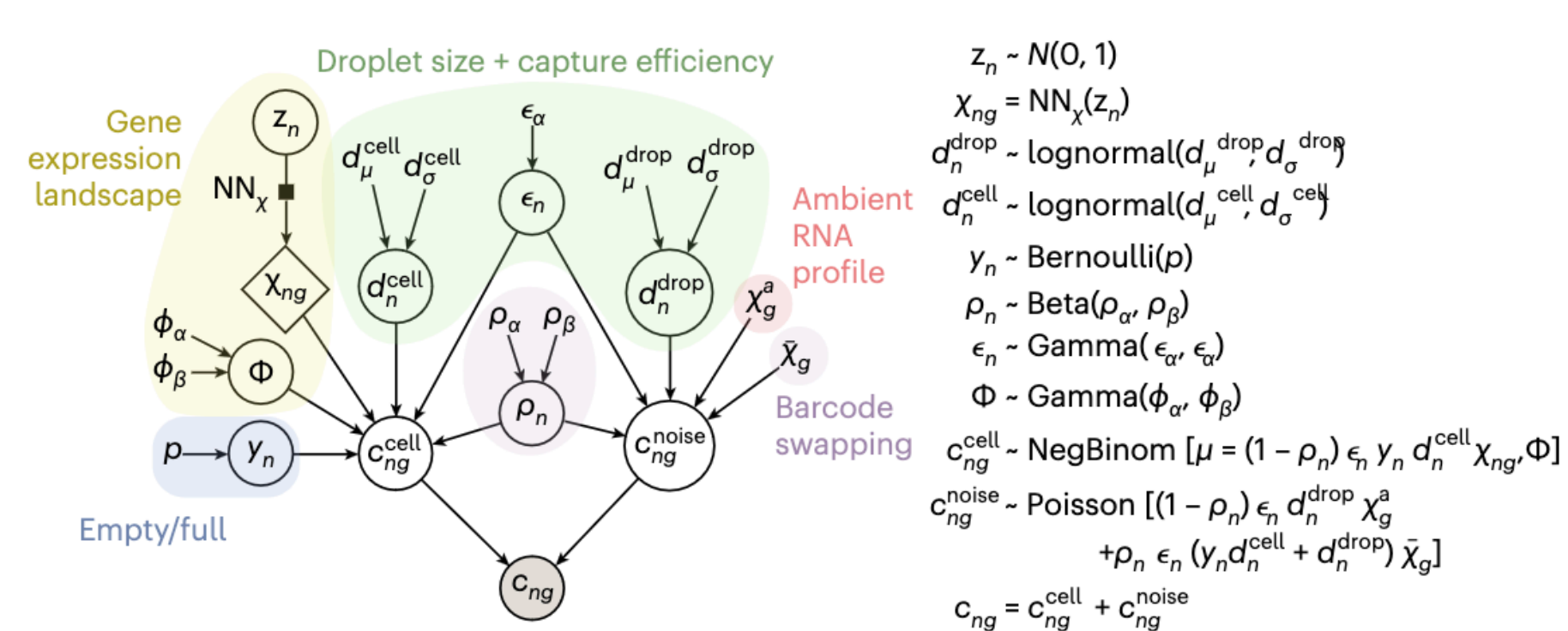
CellBender remove-background pipeline



Model schema



CellBender deep generative model



$$\begin{aligned}
 z_n &\sim N(0, 1) \\
 x_{ng} &= NN_X(z_n) \\
 d_n^{drop} &\sim \text{lognormal}(d_\mu^{drop}, d_\sigma^{drop}) \\
 d_n^{cell} &\sim \text{lognormal}(d_\mu^{cell}, d_\sigma^{cell}) \\
 y_n &\sim \text{Bernoulli}(p) \\
 \rho_n &\sim \text{Beta}(\rho_\alpha, \rho_\beta) \\
 \epsilon_n &\sim \text{Gamma}(\epsilon_\alpha, \epsilon_\beta) \\
 \Phi &\sim \text{Gamma}(\phi_\alpha, \phi_\beta) \\
 c_{ng}^{cell} &\sim \text{NegBinom}[\mu = (1 - \rho_n) \epsilon_n y_n d_n^{cell} x_{ng}, \Phi] \\
 c_{ng}^{noise} &\sim \text{Poisson}[(1 - \rho_n) \epsilon_n d_n^{drop} x_g^a + \rho_n \epsilon_n (y_n d_n^{cell} + d_n^{drop}) \bar{x}_g] \\
 c_{ng} &= c_{ng}^{cell} + c_{ng}^{noise}
 \end{aligned}$$

Overview of Cell Bender approach for empty droplet and background removal

Most work has focused on “downstream” analysis

- Imputation : detection and inference of “dropouts”*

Peng et al. *Genome Biology* (2019) 20:88
<https://doi.org/10.1186/s13059-019-1681-8>

Genome Biology

METHOD

Open Access

SCRABBLE: single-cell RNA-seq imputation constrained by bulk RNA-seq data

Tao Peng¹, Qin Zhu², Penghang Yin³ and Kai Tan^{1,2,4,5,6,7*} 

Research | [Open Access](#) | Published: 27 August 2020

A systematic evaluation of single-cell RNA-sequencing imputation methods

[Wenpin Hou](#), [Zhicheng Ji](#), [Hongkai Ji](#)  & [Stephanie C. Hicks](#) 

Genome Biology **21**, Article number: 218 (2020) | [Cite this article](#)

808 Accesses | 89 Altmetric | [Metrics](#)



ARTICLE

DOI: [10.1038/s41467-018-03405-7](https://doi.org/10.1038/s41467-018-03405-7)

OPEN

An accurate and robust imputation method scImpute for single-cell RNA-seq data

Wei Vivian Li ¹ & Jingyi Jessica Li ^{1,2}

nature **methods**

BRIEF COMMUNICATION

<https://doi.org/10.1038/s41592-018-0033-z>

SAVER: gene expression recovery for single-cell RNA sequencing

Mo Huang¹, Jingshu Wang¹, Eduardo Torre^{2,3}, Hannah Dueck⁴, Sydney Shaffer³, Roberto Bonasio⁵, John I. Murray⁴, Arjun Raj^{3,4}, Mingyao Li⁶ and Nancy R. Zhang^{1*}

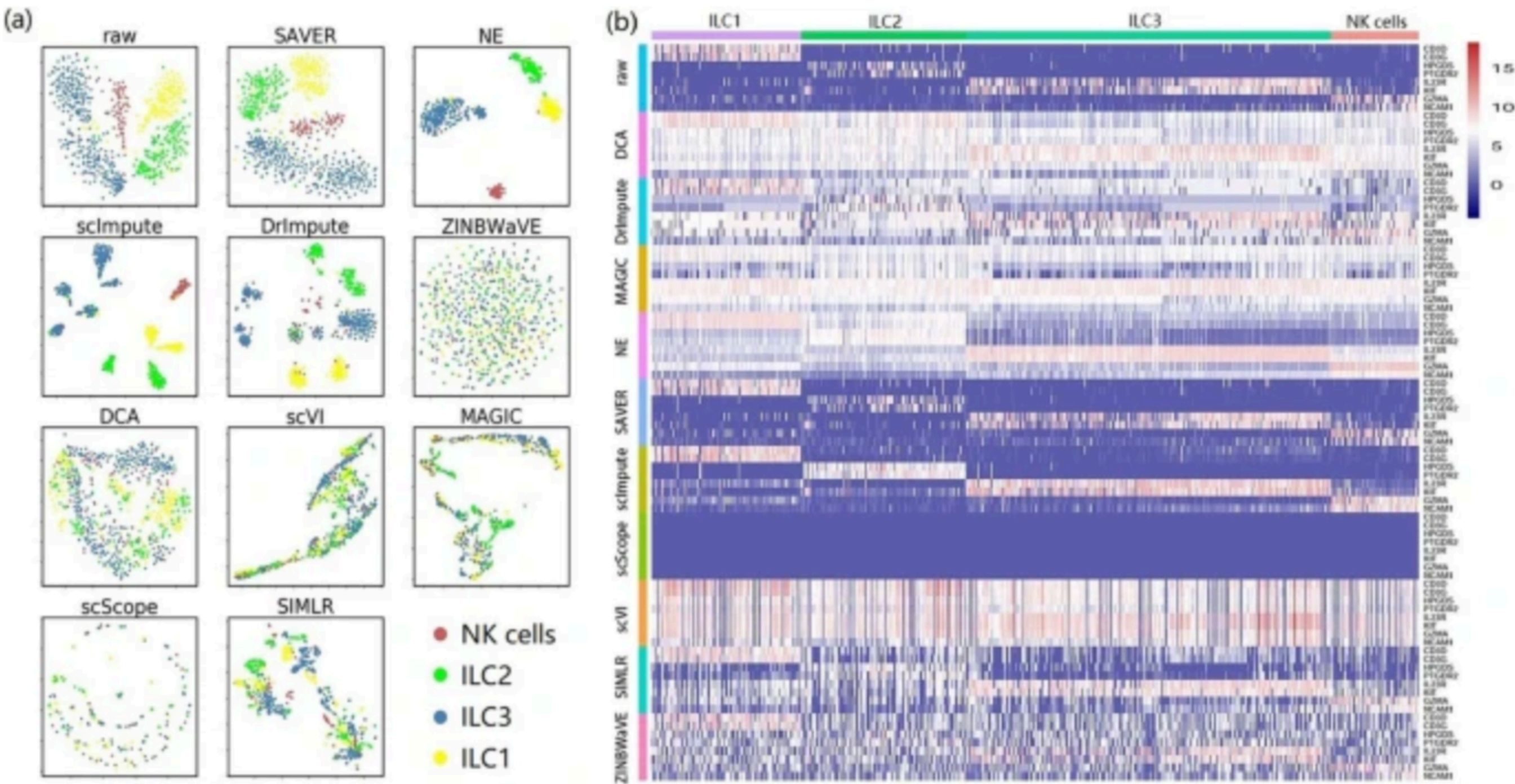
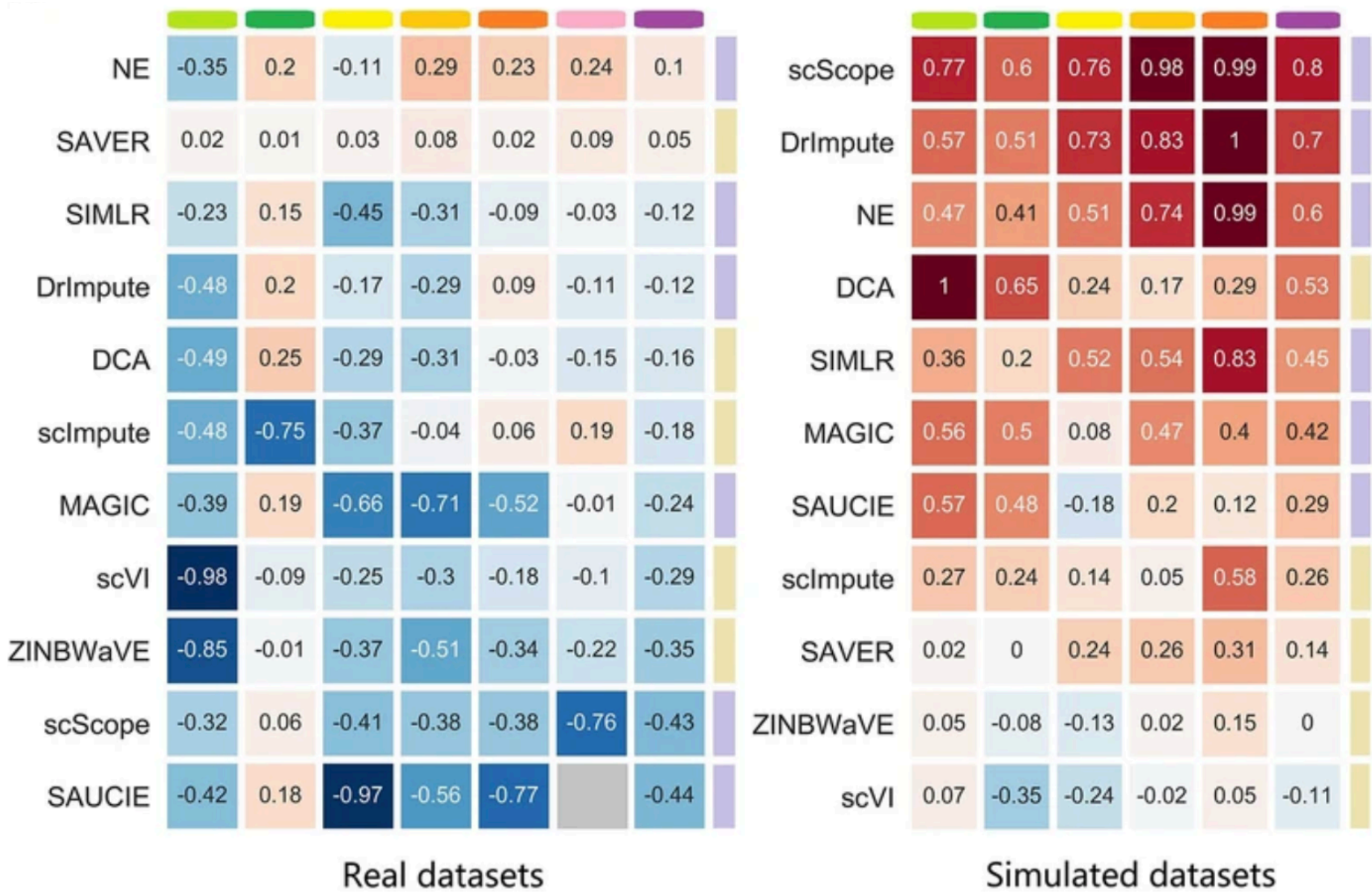
We found that the majority of scRNA-seq imputation methods outperformed no imputation in recovering gene expression observed in bulk RNA-seq. However, the majority of the methods did not improve performance in downstream analyses compared to no imputation, in particular for clustering and trajectory analysis, and thus should be used with caution.

Unclear how much imputation actually helps

Research article | [Open access](#) | Published: 28 July 2023

Evaluating imputation methods for single-cell RNA-seq data

Yi Cheng, [Xiuli Ma](#)✉, [Lang Yuan](#), [Zhaoguo Sun](#) & [Pingzhang Wang](#)✉



Performance of different imputation methods on ILC. **a** Cluster visualization by t-SNE. Colored cell labels were directly derived from the original study. **b** A heat map illustration of marker gene expression. Expression values were calculated from $\log_2(\text{raw count} + 1)$. The color bar indicates expression levels from high (red) to low (blue)

Most work has focused on “downstream” analysis

Label projection

v2.0.0

v1.0.0

Info



6 datasets



16 methods



Repository



MIT



3 control methods



4 metrics



Issues

Automated cell type annotation from rich, labeled reference data

Task info

Method info

Metric info

Dataset info

Results

A major challenge for integrating single cell datasets is creating matching cell type annotations for each cell. One of the most common strategies for annotating cell types is referred to as [“cluster-then-annotate”](#) whereby cells are aggregated into clusters based on feature similarity and then manually characterized based on differential gene expression or previously identified marker genes. Recently, methods have emerged to build on this strategy and annotate cells using [known marker genes](#). However, these strategies pose a difficulty for integrating atlas-scale datasets as the particular annotations may not match.

To ensure that the cell type labels in newly generated datasets match existing reference datasets, some methods align cells to a previously annotated [reference dataset](#) and then *project* labels from the reference to the new dataset.

Here, we compare methods for annotation based on a reference dataset. The datasets consist of two or more samples of single cell profiles that have been manually annotated with matching labels. These datasets are then split into training and test batches, and the task of each method is to train a cell type classifier on the training set and project those labels onto the test set.

Most work has focused on “downstream” analysis

- Clustering and cell-type identification : Which cells are of the same “type”? What types of cells are present in my experiment?

nature|methods

Brief Communication | Published: 27 March 2017

SC3: consensus clustering of single-cell RNA-seq data

Vladimir Yu Kiselev, Kristina Kirschner, Michael T Schaub, Tallulah Andrews, Andrew Yiu, Tamir Chandra, Kedar N Natarajan, Wolf Reik, Mauricio Barahona, Anthony R Green & Martin Hemberg 

Nature Methods **14**, 483–486 (2017) | [Download Citation](#) 

Published online 21 June 2019

Nucleic Acids Research, 2019, Vol. 47, No. 16 e95
doi: 10.1093/nar/gkz543

CHETAH: a selective, hierarchical cell type identification method for single-cell RNA sequencing

Jurrian K. de Kanter[†], Philip Lijnzaad[†], Tito Candelli, Thanasis Margaritis and Frank C.P. Holstege ^{*}

Princess Máxima Center for Pediatric Oncology, Heidelberglaan 25, 3584 CS, Utrecht, The Netherlands

Received February 22, 2019; Revised June 05, 2019; Editorial Decision June 06, 2019; Accepted June 08, 2019

Clustering and classification methods for single-cell RNA-seq data

Ren Qi , Anjun Ma, Qin Ma, Quan Zou

Briefings in Bioinformatics, bbz062, <https://doi.org/10.1093/bib/bbz062>

Published: 04 July 2019 **Article history** 

Most work has focused on “downstream” analysis

- Pseudotime analysis: What expression changes are important / determinative in development? What are developmental branchpoints? What is the Waddington landscape?

nature | **methods**

Brief Communication | Published: 21 August 2017

Reversed graph embedding resolves complex single-cell trajectories

Xiaojie Qiu, Qi Mao, Ying Tang, Li Wang, Raghav Chawla, Hannah A Pliner & Cole Trapnell 

Nature Methods **14**, 979–982 (2017) | [Download Citation](#) 

5211 Accesses | **230** Citations | **31** Altmetric | [Metrics](#) 

nature
COMMUNICATIONS

Article | [Open Access](#) | Published: 23 April 2019

Single-cell trajectories reconstruction, exploration and mapping of omics data with STREAM

Huidong Chen, Luca Albergante, Jonathan Y. Hsu, Caleb A. Lareau, Giosuè Lo Bosco, Jihong Guan, Shuigeng Zhou, Alexander N. Gorban, Daniel E. Bauer, Martin J. Aryee, David M. Langenau, Andrei Zinovyev, Jason D. Buenrostro, Guo-Cheng Yuan  & Luca Pinello 

Nature Communications **10**, Article number: 1903 (2019) | [Download Citation](#) 

8071 Accesses | **2** Citations | **44** Altmetric | [Metrics](#) 

nature
COMMUNICATIONS

Article | [Open Access](#) | Published: 22 June 2018

Uncovering pseudotemporal trajectories with covariates from single cell and bulk expression data

Kieran R Campbell & Christopher Yau 

RNA velocity

LETTER

<https://doi.org/10.1038/s41586-018-0414-6>

RNA velocity of single cells

Gioele La Manno^{1,2}, Ruslan Soldatov³, Amit Zeisel^{1,2}, Emelie Braun^{1,2}, Hannah Hochgerner^{1,2}, Viktor Petukhov^{3,4}, Katja Lidschreiber⁵, Maria E. Kastriti⁶, Peter Lönnerberg^{1,2}, Alessandro Furlan¹, Jean Fan³, Lars E. Borm^{1,2}, Zehua Liu³, David van Bruggen¹, Jimin Guo³, Xiaoling He⁷, Roger Barker⁷, Erik Sundström⁸, Gonçalo Castelo-Branco¹, Patrick Cramer^{5,9}, Igor Adameyko⁶, Sten Linnarsson^{1,2*} & Peter V. Kharchenko^{3,10*}

Based on the model of transcription shown in Fig. 1, we can write down the rate equations for a single gene, which describes how the expected number of unspliced mRNA molecules u , and spliced molecules s , evolve over time:

$$\frac{du}{dt} = \alpha(t) - \beta(t) u(t) \tag{1}$$

$$\frac{ds}{dt} = \beta(t) u(t) - \gamma(t)s(t) \tag{2}$$

Here, $\alpha(t)$ is the time-dependent rate of transcription, $\beta(t)$ is the rate of splicing, $\gamma(t)$ is the rate of degradation. Under an assumption of constant (time-independent) rates $\alpha(t) = \alpha$, $\gamma(t) = \gamma$, and setting $\beta(t) = 1$ (*i.e.* measuring all rates in units of the splicing rate), the rate equations simplify to:

$$\frac{du}{dt} = \alpha - u(t) \tag{3}$$

$$\frac{ds}{dt} = u(t) - \gamma s(t) \tag{4}$$

Uses this differential model, along with the spliced and unspliced count matrices to estimate the dynamics of each gene across cells

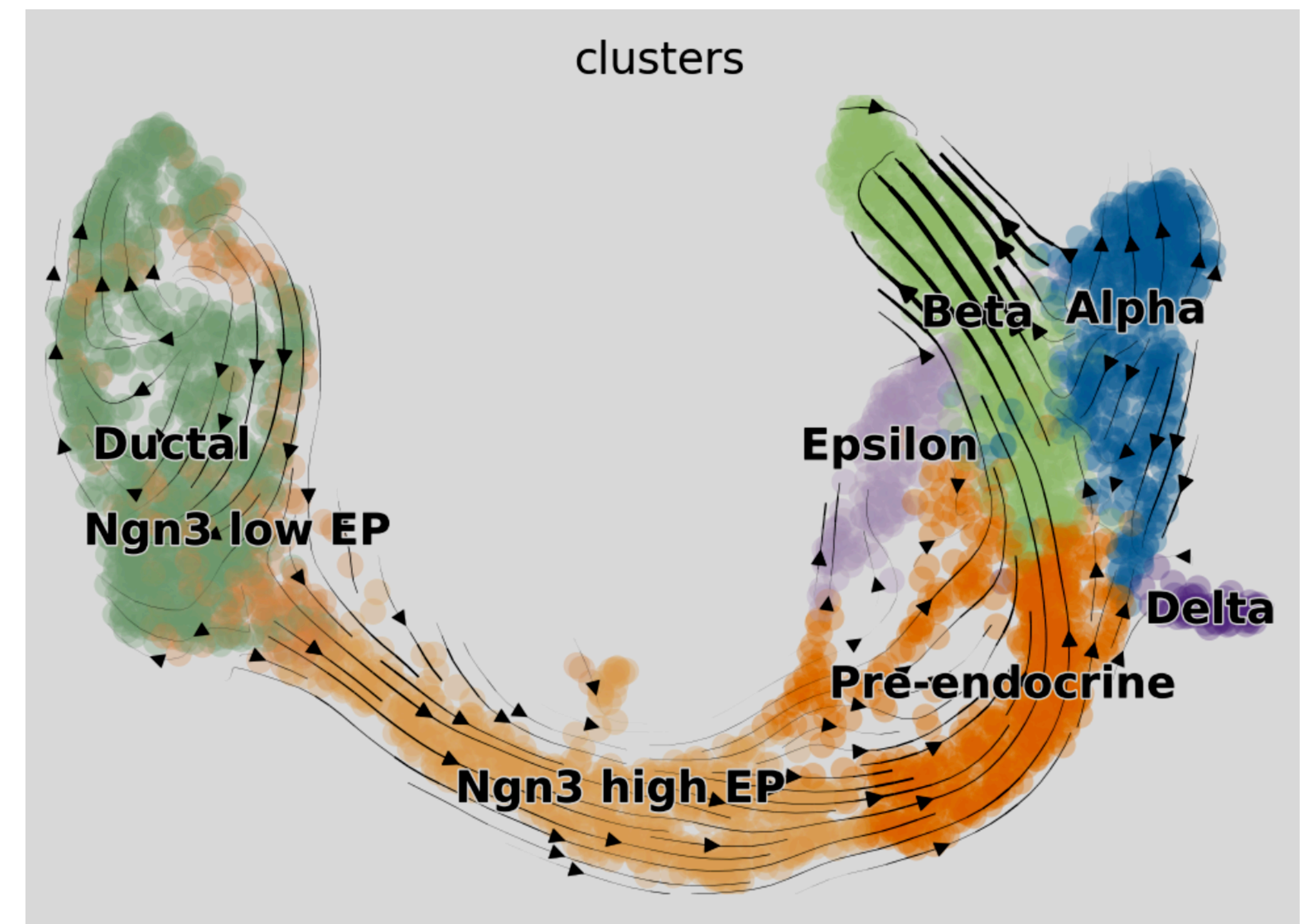
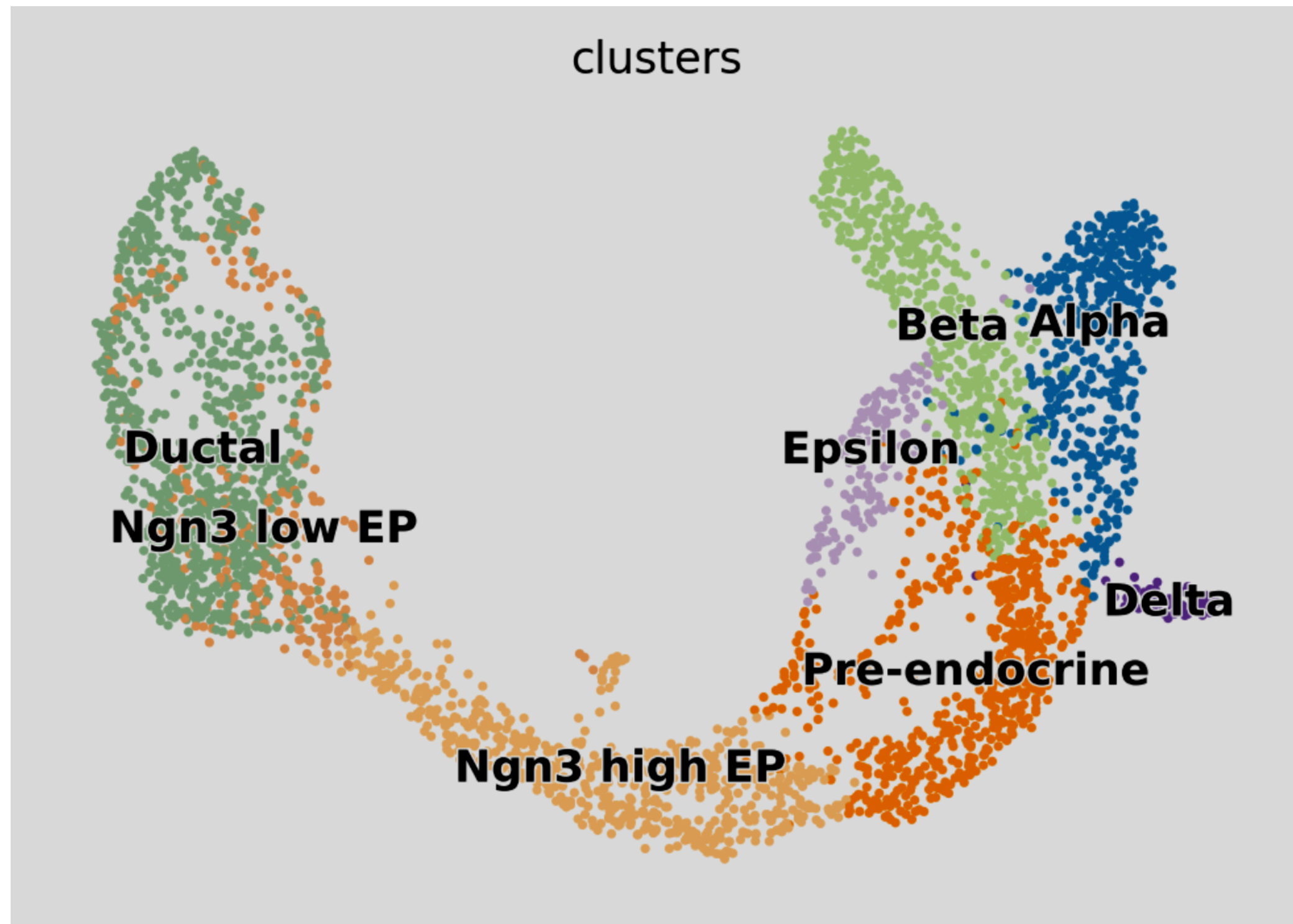
$$u(t) = \alpha(1 - e^{-t}) + u_0 e^{-t} \tag{5}$$

$$s(t) = \frac{e^{-t(1+\gamma)}[e^{t(1+\gamma)}\alpha(\gamma-1) + e^{t\gamma}(u_0-\alpha)\gamma + e^t(\alpha-\gamma(s_0+u_0+s_0\gamma))]}{\gamma(\gamma-1)} \tag{6}$$

with the initial conditions $u(0) = u_0$ and $s(0) = s_0$. This solution can be used to extrapolate mRNA abundance s to a future timepoint t_1 , under the assumption stated above, by entering the current state of the cell as u_0 and s_0 , and then computing $s(t_1)$.

This can then be used to predict how cells move through the underlying space, producing spliced RNA from unspoiled pre-cursor molecules.

RNA velocity



RNA velocity

Some concerns with RNA velocity; the assumptions and how it is actually computed

RESEARCH

Open Access

Pumping the brakes on RNA velocity by understanding and interpreting RNA velocity estimates



Shijie C. Zheng¹, Genevieve Stein-O'Brien^{2,3,4,5}, Leandros Boukas^{1,2}, Loyal A. Goff^{2,3,4*} and Kasper D. Hansen^{1,2*} 

Our findings emphasize the importance of choices in the RNA velocity workflow and highlight critical limitations of data analysis. We advise against over-interpreting expression dynamics using RNA velocity, particularly in terms of speed. Finally, we emphasize that the use of RNA velocity in assessing the correctness of a low-dimensional embedding is circular.

RESEARCH ARTICLE

RNA velocity unraveled

Gennady Gorin¹ , Meichen Fang² , Tara Chari² , Lior Pachter^{2,3*} 

¹ Division of Chemistry and Chemical Engineering, California Institute of Technology, Pasadena, California, United States of America, ² Division of Biology and Biological Engineering, California Institute of Technology, Pasadena, California, United States of America, ³ Department of Computing and Mathematical Sciences, California Institute of Technology, Pasadena, California, United States of America

Single-cell sequencing data are snapshots of biological processes, making it challenging to infer dynamic relationships between cell types. RNA velocity attempts to bypass this challenge by treating the unspliced RNA content as a proxy for spliced RNA content in the near future, and using this “extrapolation” to build directional relationships.

However, the method, as implemented in several software packages, is not yet reliable enough to be actionable, in part due to the large number of arbitrary, user-set hyperparameters, as well as fundamental incompatibilities between the biophysics of transcription in the living cell and the models used throughout the velocity workflows.

Most work has focused on “downstream” analysis

Dimensionality reduction for visualisation

v1.0.0

Reduction of high-dimensional datasets to 2D for visualization & interpretation

Task info

Method info

Metric info

Dataset info

Results

Info

4 datasets

23 methods

Repository

MIT

3 control methods

10 metrics

Issues

Dimensionality reduction is one of the key challenges in single-cell data representation. Routine single-cell RNA sequencing (scRNA-seq) experiments measure cells in roughly 20,000–30,000 dimensions (i.e., features – mostly gene transcripts but also other functional elements encoded in mRNA such as lncRNAs). Since its inception, scRNA-seq experiments have been growing in terms of the number of cells measured. Originally, cutting-edge SmartSeq experiments would yield a few hundred cells, at best. Now, it is not uncommon to see experiments that yield over [100,000 cells](#) or even [> 1 million cells](#).

Each *feature* in a dataset functions as a single dimension. While each of the ~30,000 dimensions measured in each cell contribute to an underlying data structure, the overall structure of the data is challenging to display in few dimensions due to data sparsity and the “[curse of dimensionality](#)” (distances in high dimensional data don’t distinguish data points well). Thus, we need to find a way to [dimensionally reduce](#) the data for visualization and interpretation.

Most work has focused on “downstream” analysis

- Dimensionality reduction for visualization : tSNE and UMAP are currently most popular. How do we visualize $10^3 - 10^6$ cells?

nature
biotechnology

Analysis | Published: 03 December 2018

Dimensionality reduction for visualizing single-cell data using UMAP

Etienne Becht, Leland McInnes, John Healy, Charles-Antoine Dutertre, Immanuel W H Kwok, Lai Guan Ng, Florent Gehoux & Evan W Newell 

nature
COMMUNICATIONS

Article | [Open Access](#) | Published: 21 May 2018

Interpretable dimensionality reduction of single cell transcriptome data with deep generative models

Jiarui Ding , Anne Condon & Sohrab P. Shah 

Nature Communications 9, Article number: 2002 (2018) | [Download Citation](#) 

6813 Accesses | 25 Citations | 62 Altmetric | [Metrics](#) 

Journal of Machine Learning Research 9 (2008) 2579-2605

Submitted 5/08; Revised 9/08; Published 11/08

Visualizing Data using t-SNE

Laurens van der Maaten
TiCC
Tilburg University
P.O. Box 90153, 5000 LE Tilburg, The Netherlands

LVDMAATEN@GMAIL.COM

Geoffrey Hinton
Department of Computer Science
University of Toronto
6 King's College Road, M5S 3G4 Toronto, ON, Canada

HINTON@CS.TORONTO.EDU

Hierarchical & reference-guided visualization

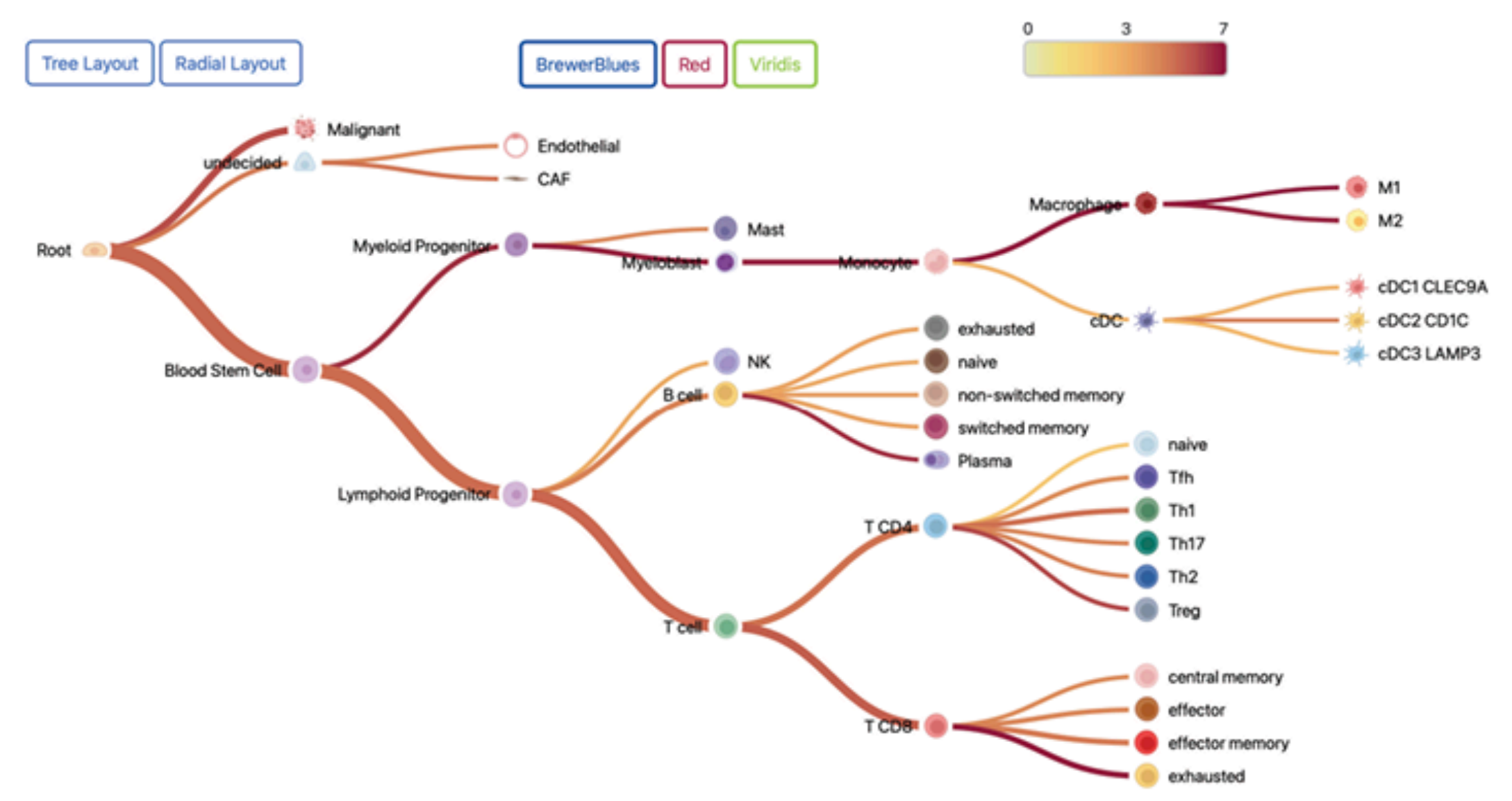
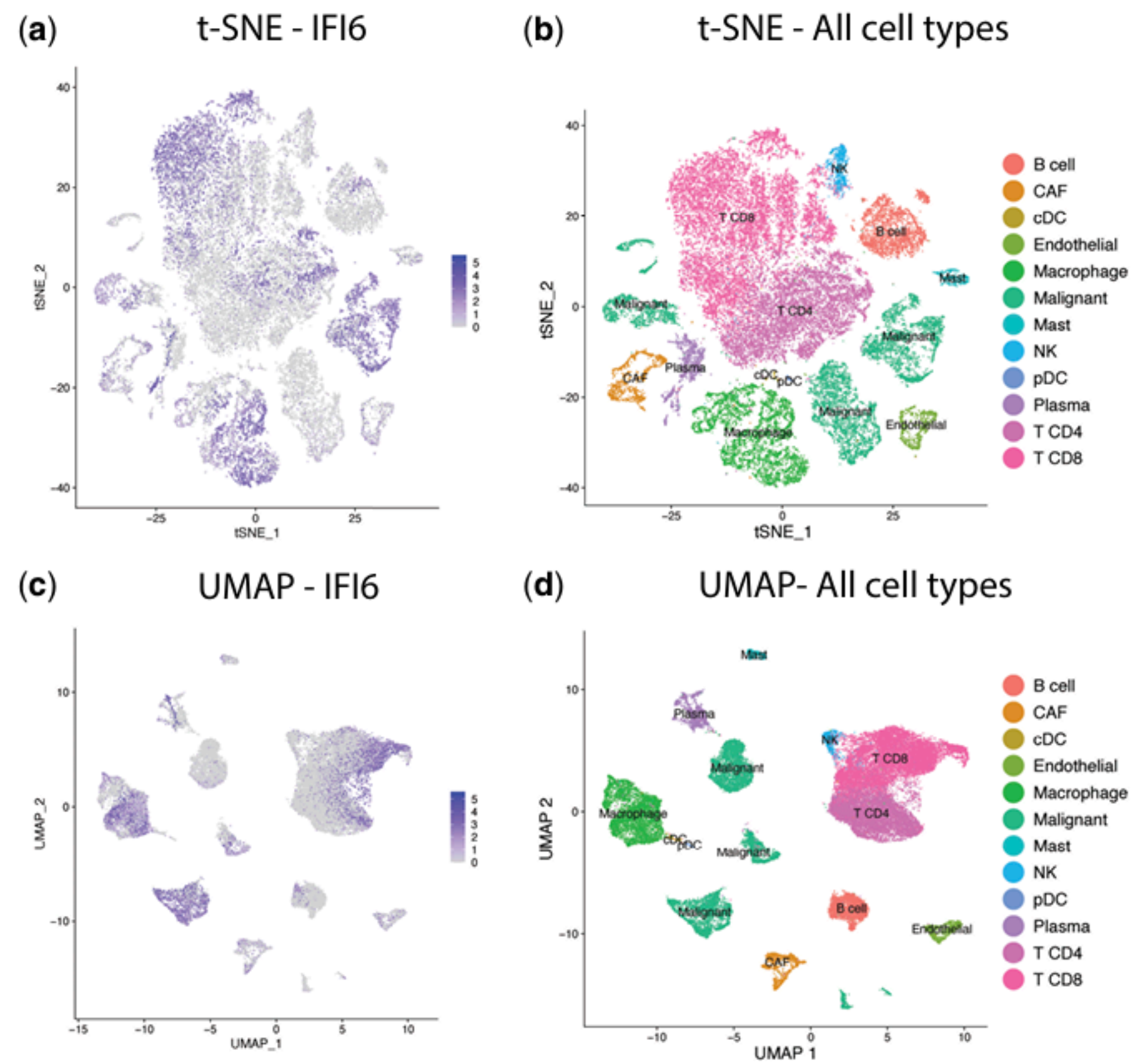


Figure 1. Visualization of *IFI6* expression across all cell types. (a) Gene expression visualized by t-SNE. (b) Cell type annotations for panel a. (c) Gene expression visualized by UMAP. (d) Cell type annotations for panel c. (e) Collapsible tree.

Bioinformatics, 2024, 40(11), btae645
<https://doi.org/10.1093/bioinformatics/btae645>
Advance Access Publication Date: 26 October 2024
Applications Note



Gene expression

Collapsible tree: interactive web app to present collapsible hierarchies

Yuan Gao^{1,2}, Rob Patro^{1,2,*}, Peng Jiang^{1,*}

Some work aims to tackle multiple such problems at once

Many variants on the same idea : try to learn a “latent” space of single-cell expression, from which many different questions can be answered.

nature|methods

ARTICLES

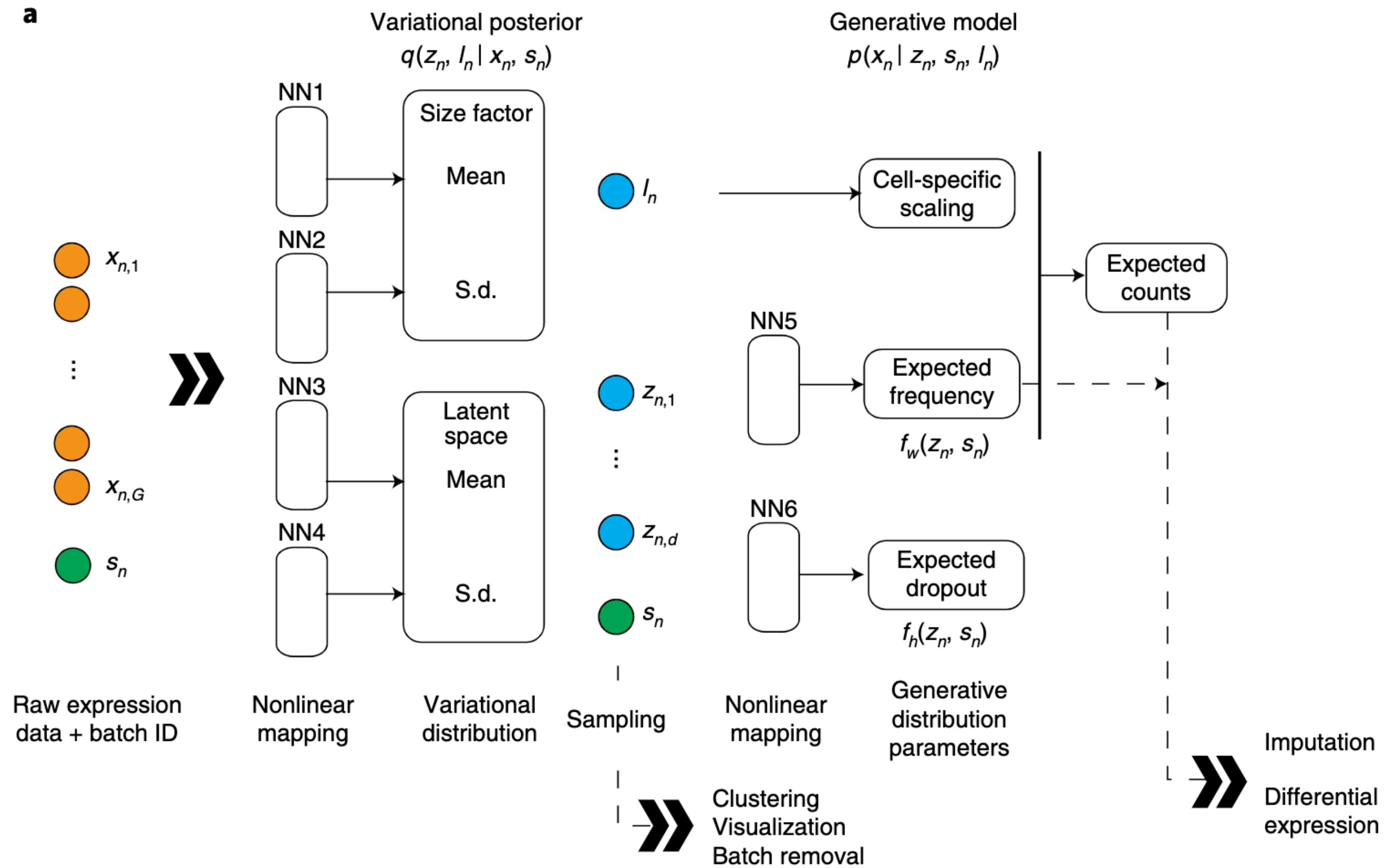
<https://doi.org/10.1038/s41592-018-0229-2>

Deep generative modeling for single-cell transcriptomics

Romain Lopez¹, Jeffrey Regier¹, Michael B. Cole², Michael I. Jordan^{1,3} and Nir Yosef^{1,4,5*}

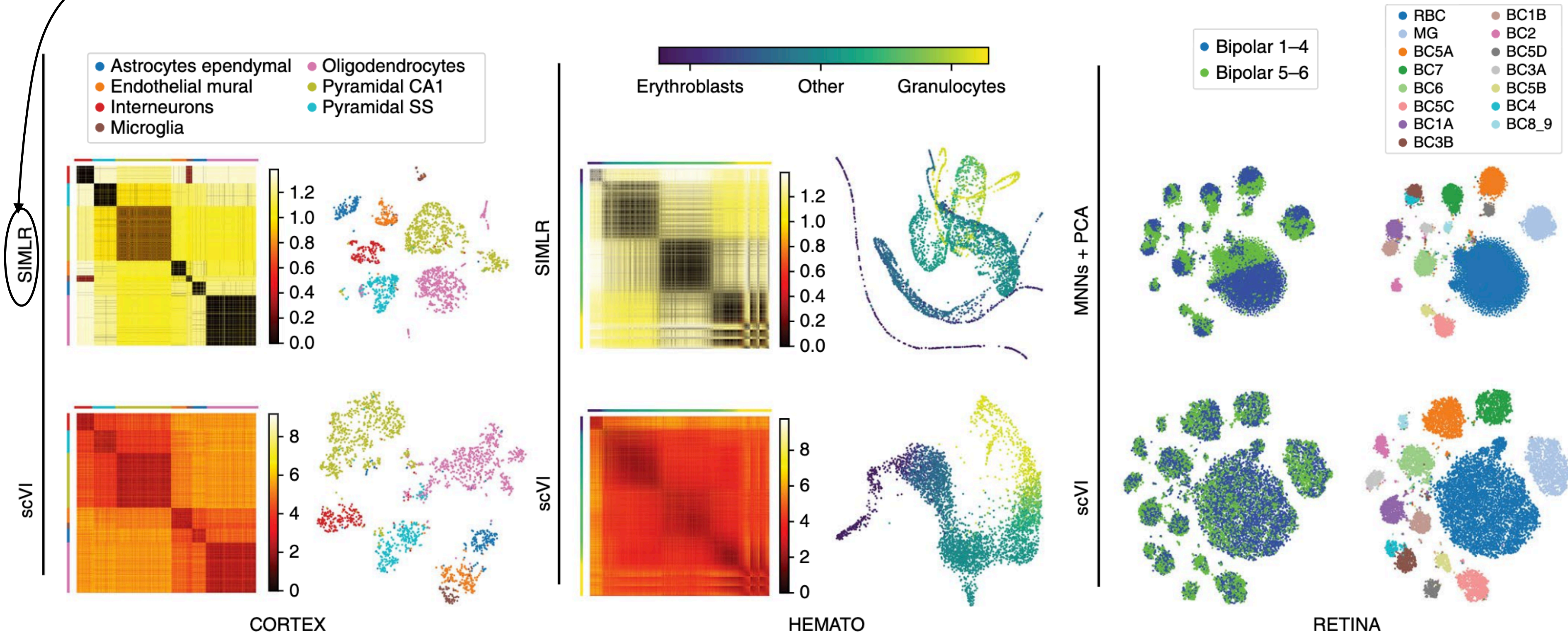
Single-cell transcriptome measurements can reveal unexplored biological diversity, but they suffer from technical noise and bias that must be modeled to account for the resulting uncertainty in downstream analyses. Here we introduce single-cell variational inference (scVI), a ready-to-use scalable framework for the probabilistic representation and analysis of gene expression in single cells (<https://github.com/YosefLab/scVI>). scVI uses stochastic optimization and deep neural networks to aggregate information across similar cells and genes and to approximate the distributions that underlie observed expression values, while accounting for batch effects and limited sensitivity. We used scVI for a range of fundamental analysis tasks including batch correction, visualization, clustering, and differential expression, and achieved high accuracy for each task.

scVI overview



scVI for clustering & vis

Method for joint clustering
and dim-reduction

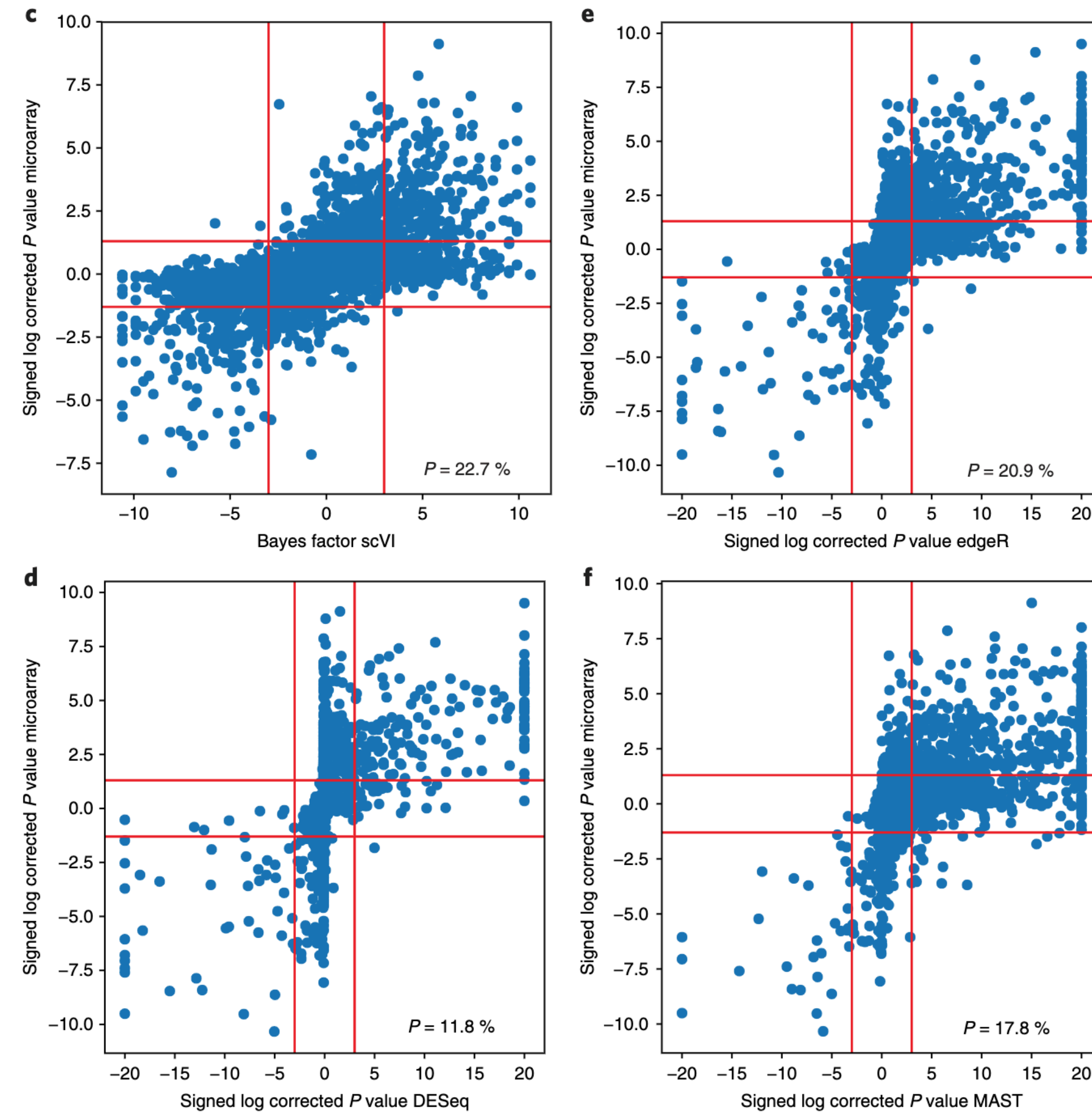


Discrete clusters

Developmental gradient

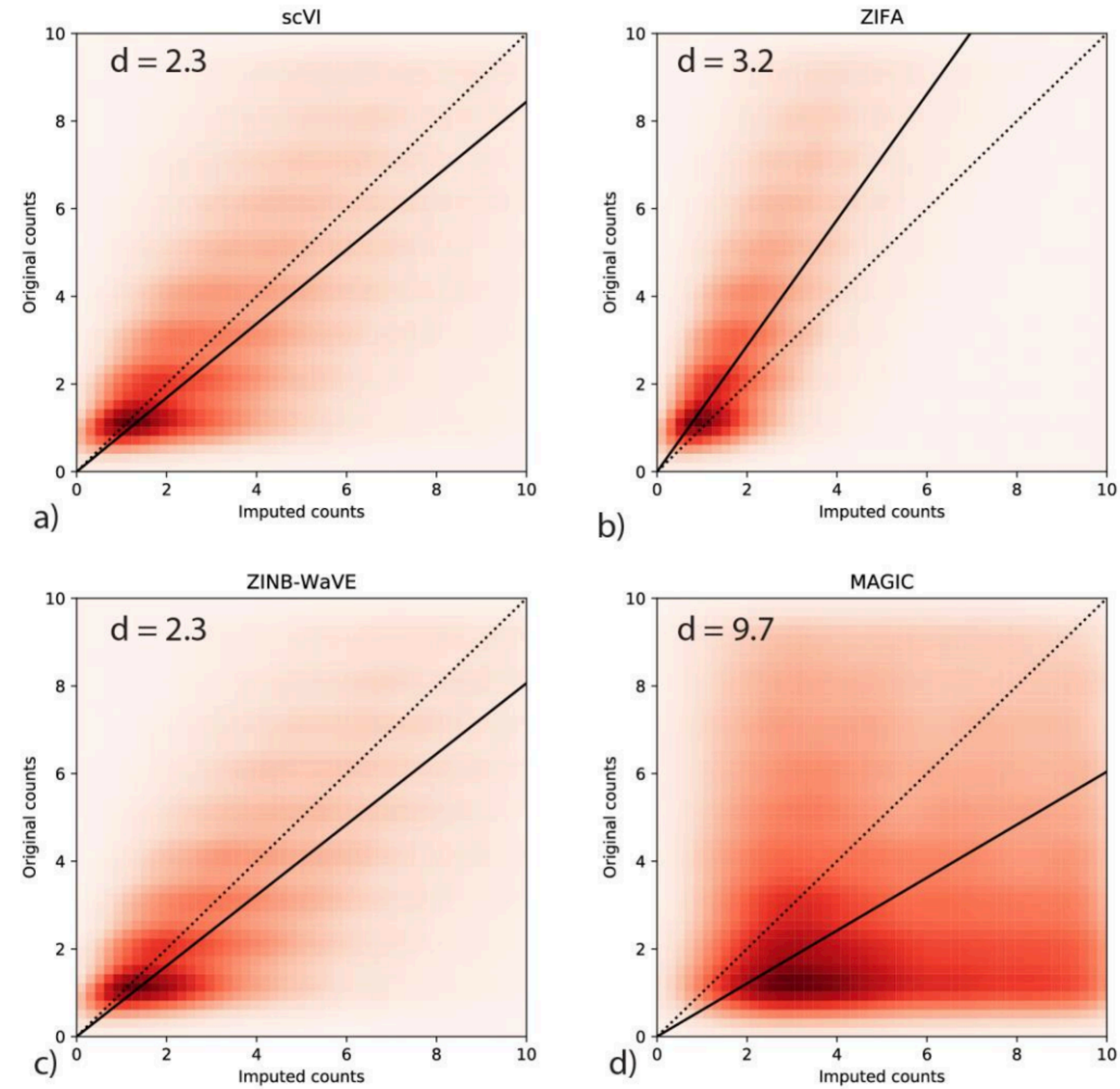
Clustering of retinal cells

scVI for differential expression



Significance levels of differential expression between B cells and dendritic cells. Points represent individual genes ($n = 3,346$). Bayes factors or BH-corrected P-values on scVI (c) compared with bulk microarray-based BH-corrected P-values. Horizontal lines denote the significance threshold of 0.05 for corrected P-values. Vertical lines denote the significance threshold for the Bayes factor of scVI (c) or 0.05 for corrected P-values for DESeq2 (d), edgeR (e), and MAST (f).

scVI for imputation



(a-d) The heat maps represent density plots of imputed values (by scVI, ZIFA, ZINB-WaVE, and MAGIC respectively) on a downsampled version versus the original (nonzero) values prior to downsampling. The reported score d is the median imputation error across all the hidden entries (lower is better; see Methods). Each density plot was computed using $n = 55,932$ independently perturbed entries from the original matrix.

scVI for batch effect removal

Batch integration

v2.0.0

v1.0.0-graph

v1.0.0-feature

v1.0.0-embed

Info



6 datasets



19 methods



Repository



MIT



7 control methods



13 metrics



Issues

Remove unwanted batch effects from scRNA-seq data while retaining biologically meaningful variation

Task info

Method info

Metric info

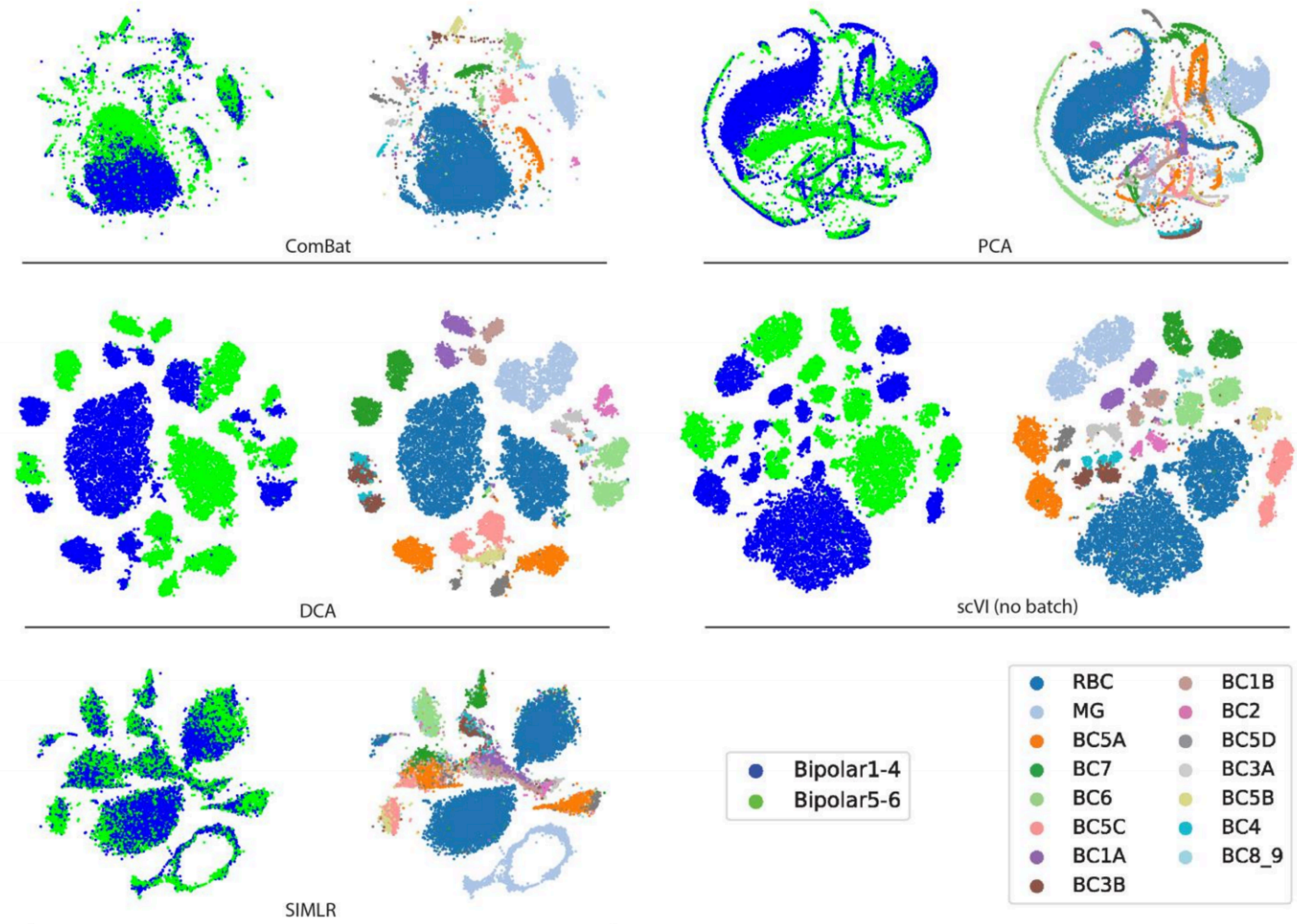
Dataset info

Results

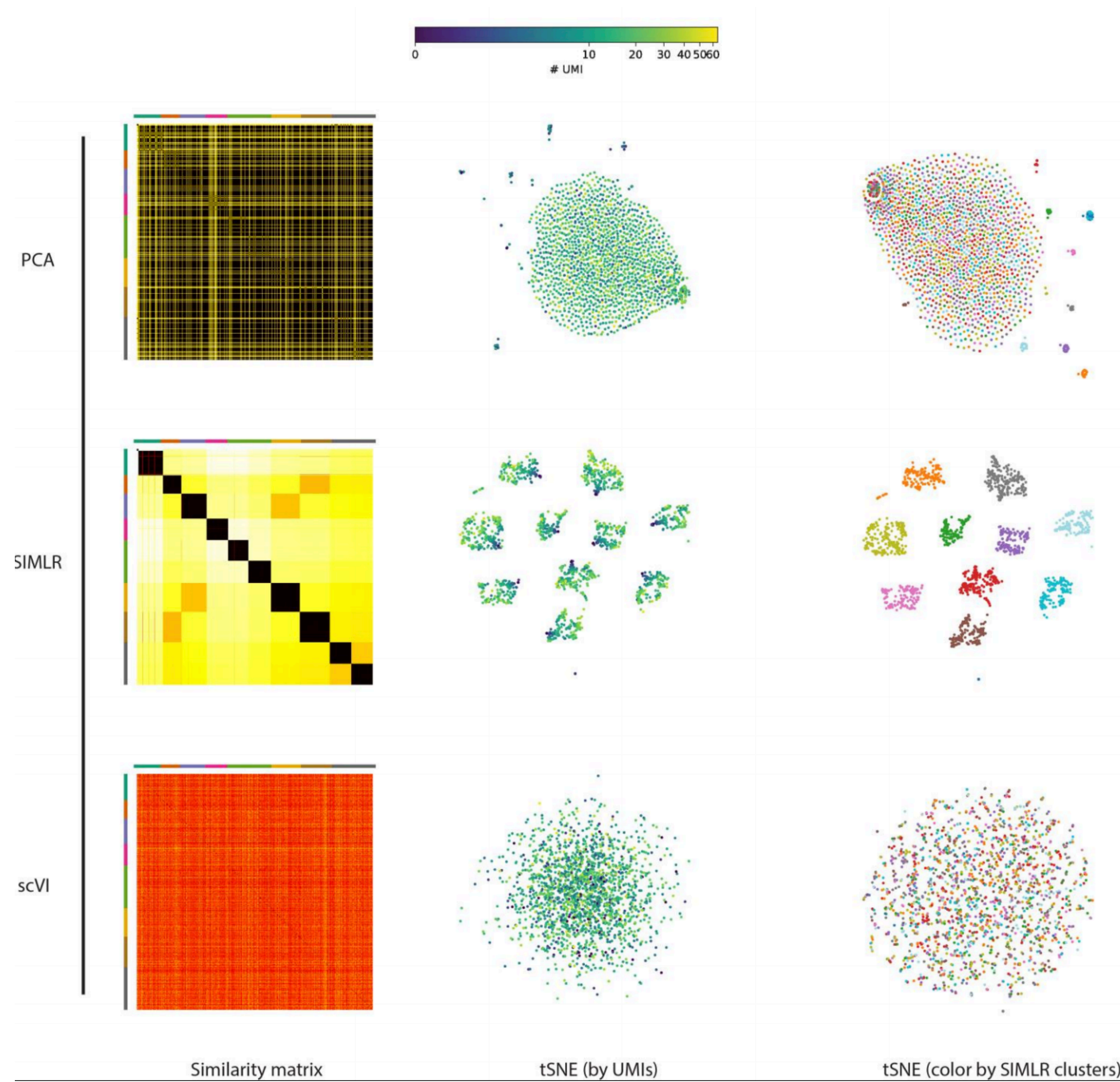
As single-cell technologies advance, single-cell datasets are growing both in size and complexity. Especially in consortia such as the Human Cell Atlas, individual studies combine data from multiple labs, each sequencing multiple individuals possibly with different technologies. This gives rise to complex batch effects in the data that must be computationally removed to perform a joint analysis. These batch integration methods must remove the batch effect while not removing relevant biological information. Currently, over 200 tools exist that aim to remove batch effects scRNA-seq datasets (Zappia et al., 2018). These methods balance the removal of batch effects with the conservation of nuanced biological information in different ways. This abundance of tools has complicated batch integration method choice, leading to several benchmarks on this topic (Luecken et al., 2021; Tran et al., 2020; Chazarra-Gil et al., 2021; Mereu et al., 2020). Yet, benchmarks use different metrics, method implementations and datasets. Here we build a living benchmarking task for batch integration methods with the vision of improving the consistency of method evaluation.

In this task we evaluate batch integration methods on their ability to remove batch effects in the data while conserving variation attributed to biological effects. As input, methods require either normalised or unnormalised data with multiple batches and consistent cell type labels. The batch integrated output can be a feature matrix, a low dimensional embedding and/or a neighbourhood graph. The respective batch-integrated representation is then evaluated using sets of metrics that capture how well batch effects are removed and whether biological variance is conserved. We have based this particular task on the latest, and most extensive benchmark of single-cell data integration methods.

scVI for batch effect removal



scVI latent representation doesn't introduce "false structure"



Foundation Models

Foundation models

v0.1.0

Modeling of single-cells to perform multiple tasks using foundation models

Task info

Method info

Metric info

Dataset info

Results

Recent developments in deep-learning have led to the creation of several 'foundation models' for single-cell data. These are large models that have been trained on data from millions of cells and aim to fully capture the variability in the single-cell landscape. Typically, they use a transformer architecture (Szałata et al., 2024) and undergo self-supervised pre-training using masking of parts of the input data. Trained foundation models can then be applied to a variety of downstream tasks, either by directly feeding new data into the model or by fine-tuning to better fit a new dataset or to produce a specific output. The general nature of single-cell foundation models and the large amount of data they have been trained on makes them potentially powerful tools for single-cell analysis but their performance is yet to be fully established.

Open Problems builds on existing evaluations (Boiarsky et al., 2023; Liu et al., 2024) of foundation models by incorporating them into our continuous benchmarking framework.

This overview combines results from the following benchmarks for individual tasks:

- [Label projection](#)
- [Batch Integration](#)

Info

 4 datasets

 8 methods

 Repository

 MIT

 2 control methods

 2 metrics

 Issues

Single-cell foundation models: bringing artificial intelligence into cell biology

Seungbyn Baek¹, Kyungwoo Song² and Insuk Lee^{1,3}✉

© The Author(s) 2025

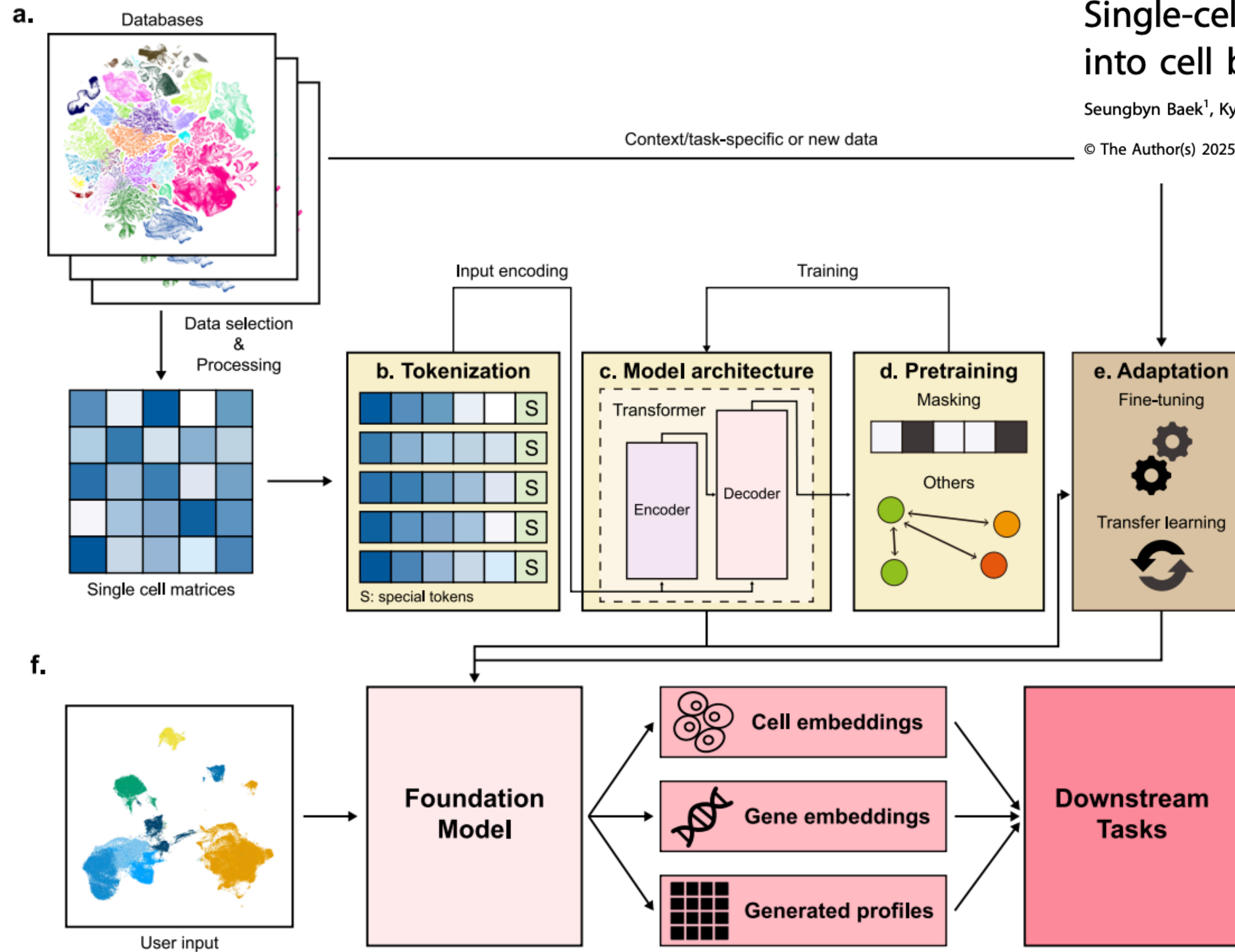


Fig. 1 Overview of scFMs. **a** Single-cell datasets are aggregated from public databases, repositories and manual searches. These datasets are processed to serve as pretraining data for scFMs, with a subset or new dataset selected to update the models to specific contexts or conditions. The example uniform manifold approximation and projection (UMAP) plots are from The Allen Brain Cell Atlas. **b**, Each cell in the pretraining datasets is converted into a model input by tokenizing its gene expressions or other measured profiles, with special tokens optionally added to denote particular cell characteristics. **c** The model architecture is built using multiple deep neural network layers, typically arranged in the transformer architecture. **d** The pretraining is carried out using various pretraining tasks, such as masking strategies and contrastive learning, among others. **e** After the initial pretraining, the model can be fine-tuned and further adapted via continual pretraining or transfer learning to meet specialized purposes or contexts. **f** The user inputs are fed into the scFMs to extract cell embeddings, gene embeddings and generated feature profiles, which can be utilized for a variety of downstream tasks.

Single-cell foundation models: bringing artificial intelligence into cell biology

Seungbyn Baek¹, Kyungwoo Song² and Insuk Lee^{1,3}✉

© The Author(s) 2025

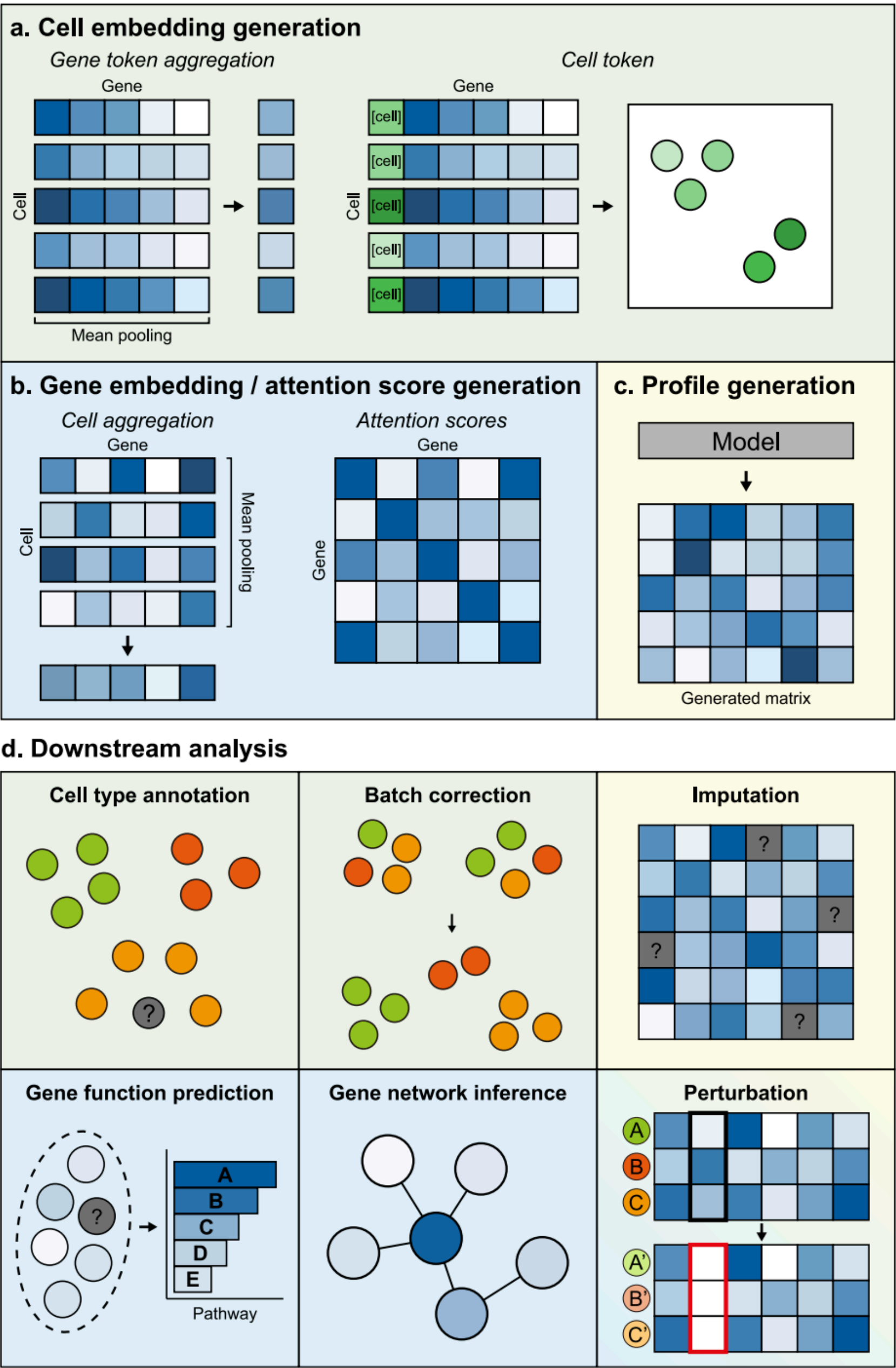


Fig. 2 Model outputs and downstream tasks. **a** The cell embeddings are derived either by aggregating gene tokens within each cell or by appending and training special tokens (cell tokens, often classification tokens) that represent individual cells. **b** The gene embeddings are produced by pooling gene-specific embeddings from all input cells. In addition, attention scores from selected layers and heads capture the relationships among genes. **c** The models generate feature profiles that represent the distribution of cells across different conditions or states. **a–c** explain initial outputs generated by feeding inputs to scFMs. **d** Various downstream tasks with scFMs. Each task is colored with types of model-generated output utilized for the task.

Single-cell foundation models: bringing artificial intelligence into cell biology

Seungbyn Baek¹, Kyungwoo Song² and Insuk Lee^{1,3}✉

© The Author(s) 2025

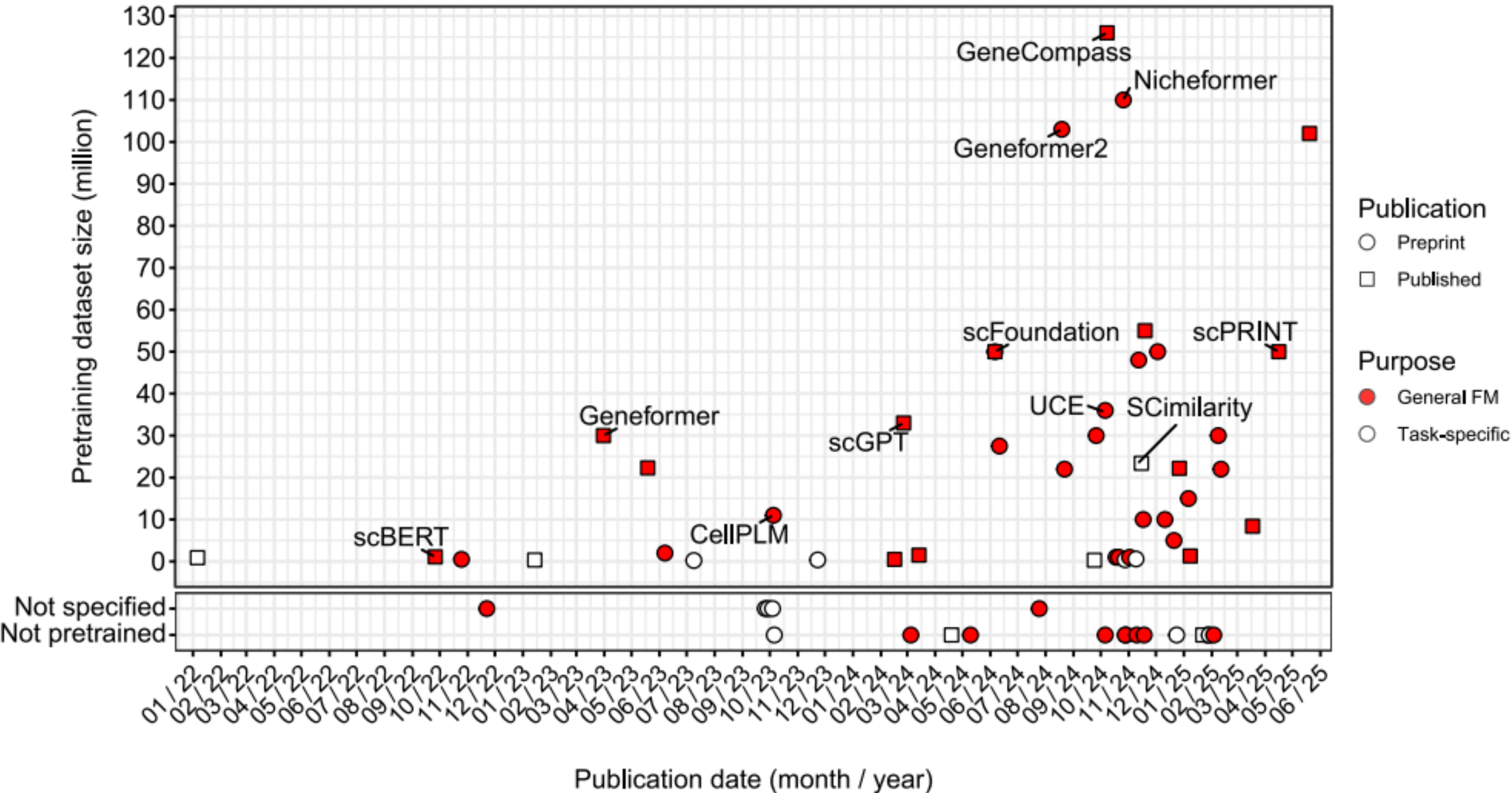


Fig. 3 Summary of recently developed scFMs. Various scFMs and task-specific models utilizing scFMs. The shape of each marker indicates its publication status as of May 2025 (for example, preprint or peer-reviewed scientific journal), whereas the color coding distinguishes the tool's purpose and characteristics, categorizing them as either general scFMs or task-specific models that leverage scFMs. The scFMs mentioned in Table 1 are labeled with their model names.

The quickly-growing repository of scRNA-seq data poses other challenges

ARTICLES

<https://doi.org/10.1038/s41592-021-01076-9>

nature | methods



Fast searches of large collections of single-cell data using scfind

Jimmy Tsz Hang Lee^{1,4}, Nikolaos Patikas^{1,2,4}, Vladimir Yu Kiselev¹ and Martin Hemberg^{1,3} ✉

Article

A cell atlas foundation model for scalable search of similar human cells

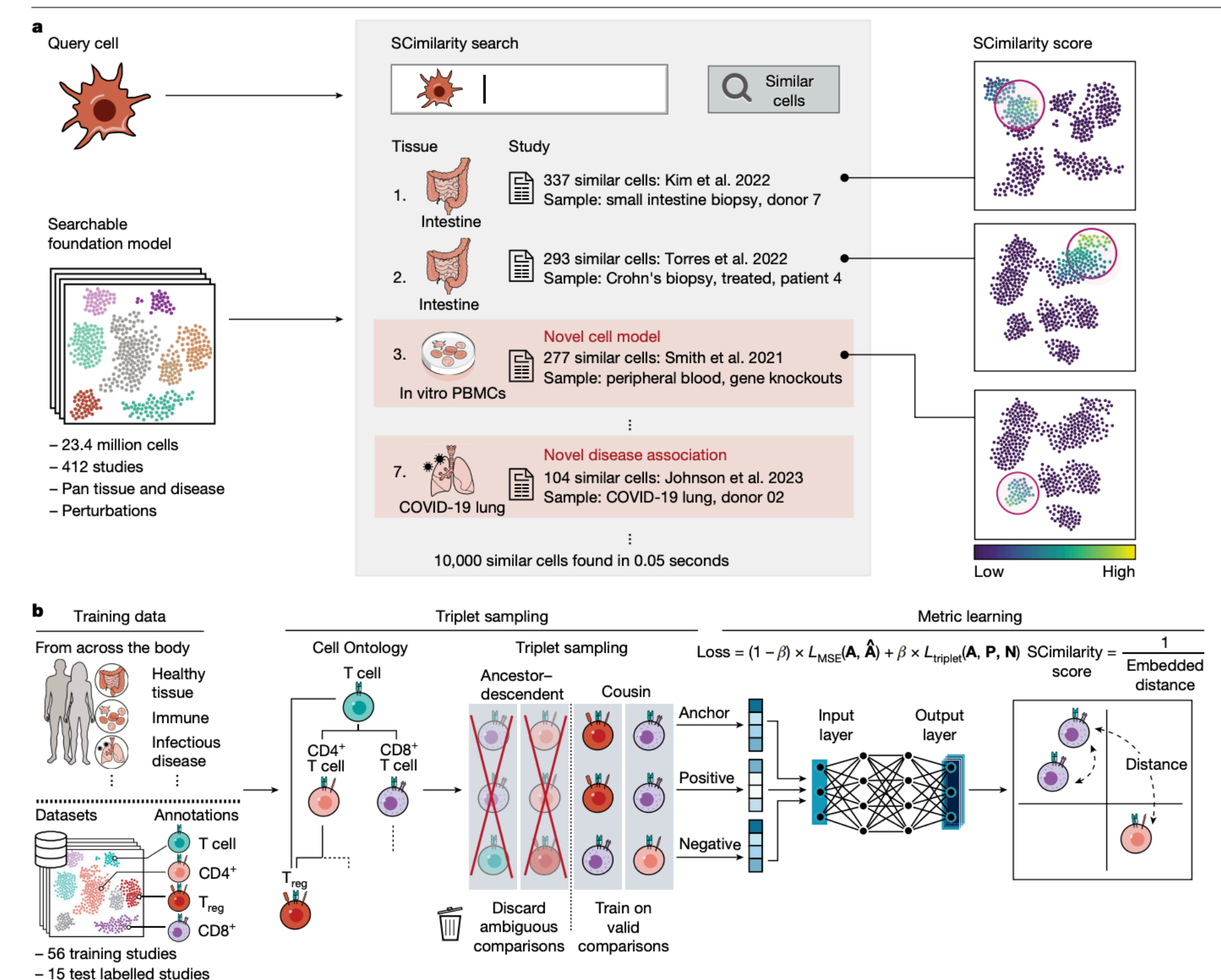
<https://doi.org/10.1038/s41586-024-08411-y>

Received: 13 August 2023

Accepted: 14 November 2024

Published online: 20 November 2024

Graham Heimberg^{1,2,7} ✉, Tony Kuo^{3,7}, Daryle J. DePianto², Omar Salem¹, Tobias Heigl², Nathaniel Diamant¹, Gabriele Scalia¹, Tommaso Biancalani¹, Shannon J. Turley^{2,4}, Jason R. Rock^{2,4}, Héctor Corrada Bravo¹, Josh Kaminker^{5,8} ✉, Jason A. Vander Heiden^{1,2,8} ✉ & Aviv Regev^{5,8} ✉



Index single-cell datasets to quickly find all cells expressing certain genes, or patterns of genes (without know which patterns to search ahead of time).

Find frequently-recurring sub-patterns as indicators of cell-type.